



Islamic University of Technology
Department of Computer Science and Engineering

Lab 4: Shortest Path Algorithm

CSE 4404: Algorithms Lab
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Task A. Shortest Path

Time Limit: 1.0 second | Memory Limit: 512 MB

You are given a weighted undirected graph with n vertices enumerated from 1 to n . Your task is to find the shortest path between vertex 1 and vertex n .

If there are multiple shortest paths, any of them will be accepted. Note that the graph may contain loops and multiple edges between the same pair of vertices.

Input Format

The first input line contains two integers n and m : the number of vertices and the number of edges ($2 \leq n \leq 10^5$, $0 \leq m \leq 10^5$).

After this, there are m lines describing the edges. Each line contains three integers a_i , b_i , and w_i : an undirected edge exists between vertex a_i and vertex b_i with weight w_i ($1 \leq a_i, b_i \leq n$, $1 \leq w_i \leq 10^6$).

Output Format

If no path exists, print -1 .

Otherwise, print the sequence of vertices representing the shortest path from vertex 1 to vertex n .

Examples

Sample Input	Sample Output
5 6 1 2 2 2 5 5 2 3 4 1 4 1 4 3 3 3 5 1	1 4 3 5
3 1 1 2 5	-1

Task B. Double Collection

Time Limit: 2 seconds | Memory Limit: 256 MB

A courier starts at city S and must deliver a package to city D . Before reaching D , the courier must collect two items: one from city A and one from city B . The items may be collected in **either order** — the courier chooses whichever sequence results in the shortest total journey.

There are N cities connected by M bidirectional roads. The i -th road connects city u_i and city v_i with a travel time of x_i . The courier always takes the fastest available route between any two cities.

Find the minimum total travel time for the courier to travel from city S to city D after collecting items from both city A and city B .

Input Format

The first line contains six integers N, M, S, D, A and B ($2 \leq N \leq 10^5, 1 \leq M \leq 3 \times 10^5, 1 \leq S, D, A, B \leq N$). It is guaranteed that S, D, A, B are all distinct.

Each of the next M lines contains three integers u_i, v_i and x_i ($1 \leq u_i, v_i \leq N, u_i \neq v_i, 1 \leq x_i \leq 10^9$) describing a bidirectional road.

It is guaranteed that a valid route always exists. There may be multiple roads between the same pair of cities.

Output Format

Output a single integer — the minimum total travel time from city S to city D after collecting items from both city A and city B .

Examples

Sample Input	Sample Output
5 6 1 5 2 4 1 3 1 3 2 1 2 4 5 3 4 2 4 5 2 1 2 6	7
5 5 1 5 3 2 1 2 1 2 3 5 1 3 2 3 4 1 4 5 3	8

Notes

In the first sample, $S = 1, D = 5, A = 2, B = 4$. Collecting A first gives the route $1 \rightarrow 2 \rightarrow 4 \rightarrow 5$ with distances $2 + 3 + 2 = 7$. Collecting B first gives $1 \rightarrow 4 \rightarrow 2 \rightarrow 5$ with distances $3 + 3 + 5 = 11$. The minimum is 7.

In the second sample, $S = 1$, $D = 5$, $A = 3$, $B = 2$. Collecting A first gives $1 \rightarrow 3 \rightarrow 2 \rightarrow 5$ with distances $2 + 3 + 7 = 12$. Collecting B first gives $1 \rightarrow 2 \rightarrow 3 \rightarrow 5$ with distances $1 + 3 + 4 = 8$. The minimum is 8.

Task C. Cheapest Fastest Route

Time Limit: 2 seconds | Memory Limit: 256 MB

A traveler wants to go from city 1 to city N as quickly as possible. However, roads have toll fees in addition to travel times, and the traveler wants to minimise the total toll paid — but only among routes that achieve the shortest possible travel time.

There are N cities connected by M bidirectional roads. The i -th road connects city u_i and city v_i with a travel time of t_i and a toll fee of c_i .

Find the minimum total toll fee among all shortest-time paths from city 1 to city N .

Input Format

The first line contains two integers N and M ($2 \leq N \leq 10^5$, $1 \leq M \leq 3 \times 10^5$).

Each of the next M lines contains four integers u_i , v_i , t_i and c_i ($1 \leq u_i, v_i \leq N$, $u_i \neq v_i$, $1 \leq t_i$, $c_i \leq 10^9$) describing a bidirectional road with travel time t_i and toll fee c_i .

It is guaranteed that city N is reachable from city 1. There may be multiple roads between the same pair of cities.

Output Format

Output a single integer — the minimum total toll fee on any shortest-time path from city 1 to city N .

Examples

Sample Input	Sample Output
4 5 1 2 3 2 1 3 2 5 2 4 2 1 3 4 3 4 2 3 1 3	3
5 6 1 2 1 5 1 3 2 1 2 4 2 3 3 4 1 4 4 5 1 2 2 5 4 1	7

Notes

In the first sample, two paths achieve the shortest travel time of 5: path $1 \rightarrow 2 \rightarrow 4$ with toll $2 + 1 = 3$, and path $1 \rightarrow 3 \rightarrow 4$ with toll $5 + 4 = 9$. The minimum toll is 3.

In the second sample, two paths achieve the shortest travel time of 4: path $1 \rightarrow 2 \rightarrow 4 \rightarrow 5$ with toll $5 + 3 + 2 = 10$, and path $1 \rightarrow 3 \rightarrow 4 \rightarrow 5$ with toll $1 + 4 + 2 = 7$. The minimum toll is 7.

Marks Distribution

Task	Marks
Task A	33%
Task B	33%
Task C	34%

Practice Problems

Problem	Links
Shortest Routes I	CSES: https://cses.fi/problemset/task/1671
Jzzhu and Train Routes	Codeforces: https://codeforces.com/problemset/problem/449/B
Dijkstra?	Codeforces: https://codeforces.com/problemset/problem/20/C
Sending Email	UVA: https://onlinejudge.org/index.php?option=onlinejudge&Itemid=8&